Course Name:	Web Development Laboratory	**Semester:**	**III**
Date of Performance:	**27/ 07/ 2026**	**DIV/ Batch No:**	**A1**
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Experiment No: 1

Title

Design and Development of an AI-based Smart Irrigation Advisory Web Portal Front-End using HTML5

Aim of the Experiment

To design and develop a multi-page, use-case-based web portal front-end using HTML5 for an AI-based Smart Irrigation Advisory System for Sugarcane Crop (KJS-AGR-01), demonstrating proper HTML5 document structure, semantic elements, formatting, lists, hyperlinks, images, image maps, tables, forms, and multimedia elements.

Objectives of the Experiment

- To develop structured and semantic HTML5 web pages for an AI-enabled smart irrigation advisory system.
- To create interactive web pages for farmer registration, irrigation recommendations, weather information, crop monitoring, and advisory services.
- To apply a real-world AI and IoT-based agriculture use case (Smart Irrigation Advisory System) as the context for front-end web development.
- To build a complete HTML5 front-end portal that can later be enhanced using CSS3, JavaScript, PHP, and React.

COs to be achieved

CO1: Developing webpages using HTML and CSS.

Books / Journals / Websites references

1. *(Students should write — e.g., MDN Web Docs HTML5 reference, W3C.)*

Theory

HTML5 is the latest version of HyperText Markup Language used to create structured and interactive web pages. It introduces semantic elements such as <header>, <nav>, <main>, <section>, <article>, and <footer>, making web pages more organized, accessible, and search-engine friendly.

HTML5 also provides support for multimedia elements like <audio> and <video>, interactive forms, tables, hyperlinks, image maps, and embedded content through <iframe>. These features help developers create user-friendly web applications for various domains such as healthcare, education, business, and agriculture.

In this experiment, HTML5 is used to design the front-end of an AI-based Smart Irrigation Advisory System. The portal enables farmers to register their farms, view irrigation schedules, monitor weather conditions, access crop information, receive AI-generated irrigation recommendations, and visualize farm locations. The application serves farmers, agricultural officers, researchers, and sugar factory management by providing organized and easy-to-understand agricultural information.

Problem Statement

"AI-based Smart Irrigation Advisory Web Portal"

An agricultural organization wants to develop a basic HTML5-based web portal front-end for an **AI-based Smart Irrigation Advisory System**. The portal should display farm information, soil moisture status, weather forecasts, irrigation recommendations, crop information, and advisory services for farmers, agricultural officers, researchers, and sugar factory management.

Each task below contributes one page or feature of this portal, representing different modules of the Smart Irrigation Advisory System.

Task 1: Create the Homepage Structure

Scenario

Design the homepage of the **Irrigation Advisory System**.

Requirements

- Use proper HTML5 document structure tags:
  - <header>, <nav>, <section>, <article>, <aside>, <footer>
- Header must contain:
  - System Name in <h1>
  - Tagline: *"Smart Irrigation using AI & Sensor Technology"*
- Navigation menu with links:
  - Home
  - Farm Dashboard

        - Irrigation Advisory
        - Sensor Data
        - Contact
- Footer with copyright information.

Concepts Tested

- HTML5 document structure
- Semantic tags
- Headings
- Hyperlinks

Deliverable: index.html

Task 2: Display Farm Information using Lists

Scenario

Display important farm information and irrigation resources.

Requirements

Create:

- Ordered List for
    - Irrigation Workflow
- Unordered List for
    - Sensors Used
        - Soil Moisture Sensor
        - Temperature Sensor
        - Humidity Sensor
        - Rainfall Sensor
- Description List for
    - Crop Stages
        - Germination
        - Tillering
        - Grand Growth
        - Maturity

Use formatting tags:

- <b>
- <i>
- <mark>

Concepts Tested

- Lists
- Formatting tags
- Text formatting

Deliverable: farm-info.html

Task 3: Irrigation Advisory Page using Formatting Tags

Scenario

Display today's irrigation recommendation for a sugarcane field.

Requirements

Display:

- Plot ID
- Farmer Name
- Soil Moisture
- Recommended Irrigation Duration
- Water Stress Level

Use:

- <strong>
- <em>
- <mark>
- <small>
-
- <hr>

Concepts Tested

- Text formatting
- Block-level elements
- Inline formatting

Deliverable: advisory.html

Task 4: Farm Gallery using Images

Scenario

Create a gallery showing different farm conditions.

Requirements

Display four images:

- Healthy Sugarcane Crop
- Drip Irrigation System
- Soil Moisture Sensor
- Weather Monitoring Station

Each image should:

- Open detailed information when clicked
- Include:
    - alt
    - title
    - width
    - height

Use:

- <figure>
- <figcaption>

Concepts Tested

- Images
- Hyperlinks
- Global attributes

Deliverable: gallery.html

Task 5: Farm Plot Navigation using Image Map

Scenario

Provide navigation to different farm plots.

Requirements

Display a farm map.

Create clickable areas for:

- Plot A
- Plot B
- Plot C

Each area should open the corresponding plot details.

Use:

- <map>
- <area>

Concepts Tested

- Image Maps
- Hyperlinks

Deliverable: plot-map.html

Task 6: Soil Moisture & Irrigation Table

Scenario

Display irrigation recommendations for multiple plots.

Requirements

Create a table with columns:

- Plot ID
- Farmer Name
- Soil Moisture (%)
- Weather
- Irrigation Recommendation
- Water Requirement

Use

- <caption>
- <thead>
- <tbody>
- <tfoot>
- rowspan
- colspan

Apply suitable formatting.

Concepts Tested

- HTML Tables
- Table Attributes

Deliverable: irrigation-table.html

Task 7: Farmer Registration Form

Scenario

Create a registration page for farmers.

Requirements

Use form controls:

- Name
- Mobile Number
- Email
- Plot ID
- Village
- Crop Stage
- Soil Type
- Irrigation Method
- Date
- File Upload (Farm Photo)

Include

- <label>
- <fieldset>
- <legend>

Use HTML5 attributes

- required
- placeholder
- pattern
- autofocus

Concepts Tested

- HTML Forms
- HTML5 Input Types
- Validation Attributes

Deliverable: registration.html

Task 8: Display Weather Forecast using IFrame

Scenario

Display weather information without leaving the website.

Requirements

Embed

- IMD Weather Forecast page
- Google Maps showing the farm location

Use

- <iframe>

Place inside a section called

Today's Weather Forecast

Concepts Tested

- Inline Frames
- External Content Integration

Deliverable: weather.html

Task 9: Add Multimedia & Executable Content

Scenario

Create a multimedia awareness page for farmers.

Requirements

Add

- A video explaining Smart Irrigation
- Background audio explaining water conservation
- JavaScript alert

Welcome to the Irrigation Advisory System

Use

- <video>
- <audio>
- <script>

Enable

- controls
- autoplay
- loop

Concepts Tested

- Multimedia Tags

- JavaScript
- Executable Content

Deliverable: media.html

Code :
Task 1

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Smart Irrigation Advisory System - Home</title>
</head>
<body>

<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>

<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>
```

```
<section>
  <article>
    <h2>About this System</h2>
    <p>This is a smart irrigation advisory portal made for sugarcane farmers.
    It uses soil moisture, temperature, humidity and rainfall sensors to
    suggest when and how much to irrigate. This helps save water and
    increase crop yield.</p>
  </article>

  <aside>
    <h3>Other Pages</h3>
    <ul>
      <li><a href="farm-info.html">Farm Information</a></li>
      <li><a href="advisory.html">Today's Advisory</a></li>
      <li><a href="gallery.html">Photo Gallery</a></li>
      <li><a href="plot-map.html">Plot Map</a></li>
      <li><a href="irrigation-table.html">Irrigation Table</a></li>
      <li><a href="registration.html">Farmer Registration</a></li>
      <li><a href="weather.html">Weather Forecast</a></li>
      <li><a href="media.html">Awareness Video</a></li>
    </ul>
  </aside>
</section>

<section id="contact">
  <h2>Contact</h2>
```

```
  <p>Department of Computer Engineering<br>
  K. J. Somaiya School of Engineering, Mumbai-77</p>
</section>

<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 2

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Farm Information</title>
</head>
<body>

<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>

<nav>
  <a href="index.html">Home</a> |
```

```html
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>

<section>
  <h2>Irrigation Workflow</h2>
  <ol>
    <li>Sensors collect data from the field.</li>
    <li>Data is sent to the AI advisory engine.</li>
    <li>The system checks <b>soil moisture</b>, weather and crop stage.</li>
    <li>The system gives an <i>irrigation recommendation</i>.</li>
    <li>The farmer is informed about the recommendation.</li>
    <li>Farmer irrigates the field.</li>
  </ol>
</section>

<section>
  <h2>Sensors Used</h2>
  <ul>
    <li><b>Soil Moisture Sensor</b></li>
    <li><b>Temperature Sensor</b></li>
    <li><b>Humidity Sensor</b></li>
    <li><mark>Rainfall Sensor</mark></li>
  </ul>
</section>
```

```
<section>
  <h2>Crop Stages</h2>
  <dl>
    <dt><b>Germination</b></dt>
    <dd>First stage after planting.</dd>

    <dt><b>Tillering</b></dt>
    <dd>Side shoots start forming.</dd>

    <dt><b>Grand Growth</b></dt>
    <dd><mark>Highest water need</mark> stage.</dd>

    <dt><b>Maturity</b></dt>
    <dd>Sugar content increases, less water needed.</dd>
  </dl>
</section>

<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 3

```
<!DOCTYPE html>
<html lang="en">
```

```html
<head>
<meta charset="UTF-8">
<title>Irrigation Advisory</title>
</head>
<body>

<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>

<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>

<section>
  <h2>Today's Irrigation Recommendation</h2>

  <p><strong>Plot ID:</strong> PLT-A-07<br>
  <strong>Farmer Name:</strong> Ramesh Patil<br>
  <strong>Soil Moisture:</strong> <mark>28%</mark> <small>(measured at 6:00
AM)</small></p>
```

```
  <hr>


  <p><strong>Recommended Irrigation Duration:</strong> <em>45 minutes</em> (drip
irrigation)<br>
  <strong>Water Stress Level:</strong> <mark>Moderate</mark></p>


</section>


<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>


</body>
</html>
```

Task 4

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Farm Gallery</title>
</head>
<body>


<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
```

```
</header>


<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>


<section>
  <h2>Farm Photo Gallery</h2>


  <figure>
    <a href="#detail-crop">
      <img src="images/crop-sugarcane.png" alt="Healthy sugarcane crop"
           title="Healthy Sugarcane Crop" width="300" height="200">
    </a>
    <figcaption>Healthy Sugarcane Crop</figcaption>
  </figure>


  <figure>
    <a href="#detail-drip">
      <img src="images/drip-irrigation.png" alt="Drip irrigation system"
           title="Drip Irrigation System" width="300" height="200">
    </a>
    <figcaption>Drip Irrigation System</figcaption>
```

```
  </figure>


  <figure>
    <a href="#detail-soil">
      <img src="images/soil-sensor.png" alt="Soil moisture sensor"
           title="Soil Moisture Sensor" width="300" height="200">
    </a>
    <figcaption>Soil Moisture Sensor</figcaption>
  </figure>


  <figure>
    <a href="#detail-weather">
      <img src="images/weather-station.png" alt="Weather monitoring station"
           title="Weather Monitoring Station" width="300" height="200">
    </a>
    <figcaption>Weather Monitoring Station</figcaption>
  </figure>
</section>


<section id="detail-crop">
  <h3>Healthy Sugarcane Crop</h3>
  <p>Sugarcane in the Grand Growth stage.</p>
</section>


<section id="detail-drip">
  <h3>Drip Irrigation System</h3>
  <p>Water is given directly at the root of the plant.</p>
```

```
</section>

<section id="detail-soil">
  <h3>Soil Moisture Sensor</h3>
  <p>Sensor placed in soil to check moisture level.</p>
</section>

<section id="detail-weather">
  <h3>Weather Monitoring Station</h3>
  <p>Records temperature, humidity and rainfall.</p>
</section>

<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 5

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Farm Plot Map</title>
</head>
<body>
```

```html
<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>


<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>


<section>
  <h2>Farm Plot Map</h2>
  <p>Click on a plot to see its details.</p>

  <img src="images/farm-map.png" alt="Farm map with Plot A, Plot B and Plot C"
       usemap="#farmmap" width="600" height="400">

  <map name="farmmap">
    <area shape="rect" coords="30,30,280,180" href="#plotA" alt="Plot A">
    <area shape="rect" coords="320,30,570,180" href="#plotB" alt="Plot B">
    <area shape="rect" coords="30,220,570,370" href="#plotC" alt="Plot C">
  </map>
</section>
```

```
<section id="plotA">
  <h3>Plot A</h3>
  <p>Area: 2.1 hectares, Crop Stage: Tillering, Soil Moisture: 34%</p>
</section>

<section id="plotB">
  <h3>Plot B</h3>
  <p>Area: 1.8 hectares, Crop Stage: Grand Growth, Soil Moisture: 22%</p>
</section>

<section id="plotC">
  <h3>Plot C</h3>
  <p>Area: 3.4 hectares, Crop Stage: Maturity, Soil Moisture: 41%</p>
</section>

<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 6

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```html
<meta charset="UTF-8">
<title>Irrigation Table</title>
</head>
<body>

<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>

<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>

<section>
  <h2>Irrigation Recommendations for Multiple Plots</h2>

  <table border="1" cellpadding="6" cellspacing="0">
    <caption>Soil Moisture and Irrigation Recommendation - 07 Aug 2026</caption>
    <thead>
      <tr>
        <th>Plot ID</th>
        <th>Farmer Name</th>
```

```html
      <th>Soil Moisture (%)</th>
      <th>Weather</th>
      <th>Irrigation Recommendation</th>
      <th>Water Requirement</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>PLT-A-07</td>
      <td>Ramesh Patil</td>
      <td>28</td>
      <td>Sunny</td>
      <td>Irrigate within 6 hours</td>
      <td>45 min (drip)</td>
    </tr>
    <tr>
      <td>PLT-B-03</td>
      <td>Suresh Jadhav</td>
      <td>22</td>
      <td>Sunny</td>
      <td rowspan="2">Irrigate immediately</td>
      <td>60 min (drip)</td>
    </tr>
    <tr>
      <td>PLT-B-04</td>
      <td>Suresh Jadhav</td>
      <td>20</td>
```

```
      <td>Sunny</td>
      <td>65 min (drip)</td>
    </tr>
    <tr>
      <td>PLT-C-11</td>
      <td>Anita More</td>
      <td>41</td>
      <td>Cloudy</td>
      <td>No irrigation needed</td>
      <td>0 min</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="5">Average Soil Moisture</td>
      <td>27.75%</td>
    </tr>
  </tfoot>
 </table>
</section>

<footer>
 <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 7

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Farmer Registration</title>
</head>
<body>

<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>

<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>

<section>
  <h2>Farmer Registration Form</h2>
```

```
<form action="#" method="post">

  <fieldset>
    <legend>Personal Details</legend>

    <label for="name">Name:</label><br>
    <input type="text" id="name" name="name" placeholder="Enter your name" required autofocus><br><br>

    <label for="mobile">Mobile Number:</label><br>
    <input type="tel" id="mobile" name="mobile" placeholder="10 digit number" pattern="[0-9]{10}" required><br><br>

    <label for="email">Email:</label><br>
    <input type="email" id="email" name="email" placeholder="you@example.com" required><br><br>
  </fieldset>

  <fieldset>
    <legend>Farm Details</legend>

    <label for="plotid">Plot ID:</label><br>
    <input type="text" id="plotid" name="plotid" placeholder="e.g. PLT-A-07" required><br><br>

    <label for="village">Village:</label><br>
    <input type="text" id="village" name="village" placeholder="Enter village name" required><br><br>
```

```
<label for="cropstage">Crop Stage:</label><br>
<select id="cropstage" name="cropstage" required>
  <option value="">--Select--</option>
  <option value="germination">Germination</option>
  <option value="tillering">Tillering</option>
  <option value="grand-growth">Grand Growth</option>
  <option value="maturity">Maturity</option>
</select><br><br>

<label for="soiltype">Soil Type:</label><br>
<select id="soiltype" name="soiltype" required>
  <option value="">--Select--</option>
  <option value="clay">Clay</option>
  <option value="loamy">Loamy</option>
  <option value="sandy">Sandy</option>
</select><br><br>

<label for="irrigation-method">Irrigation Method:</label><br>
<select id="irrigation-method" name="irrigation-method" required>
  <option value="">--Select--</option>
  <option value="drip">Drip Irrigation</option>
  <option value="sprinkler">Sprinkler</option>
  <option value="flood">Flood/Furrow</option>
</select><br><br>

<label for="regdate">Date:</label><br>
<input type="date" id="regdate" name="regdate" required><br><br>
```

```
    <label for="farmphoto">Upload Farm Photo:</label><br>
    <input type="file" id="farmphoto" name="farmphoto" accept="image/*"><br><br>
  </fieldset>

  <input type="submit" value="Register">

 </form>
</section>

<footer>
 <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>

</body>
</html>
```

Task 8

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Weather Forecast</title>
</head>
<body>

<header>
```

```
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>


<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>


<section>
  <h2>Today's Weather Forecast</h2>


  <h3>Weather Forecast</h3>
  <iframe
src="https://embed.windy.com/embed2.html?lat=19.076&lon=72.878&detailLat=19.076&detailLon=72.878&width=650&height=450&zoom=8&level=surface&overlay=wind&menu=&message=&marker=true&calendar=now&pressure=&type=map&location=coordinates&detail=&metricWind=default&metricTemp=default&radarRange=-1"
        width="600" height="450" frameborder="0">
  </iframe>


  <h3>Farm Location</h3>
  <iframe
src="https://www.openstreetmap.org/export/embed.html?bbox=72.75%2C18.98%2C72.95%2C19.15&layer=mapnik&marker=19.076%2C72.878"
        width="600" height="350" frameborder="0">
```

```
  </iframe>
</section>


<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>


</body>
</html>
```

Task 9

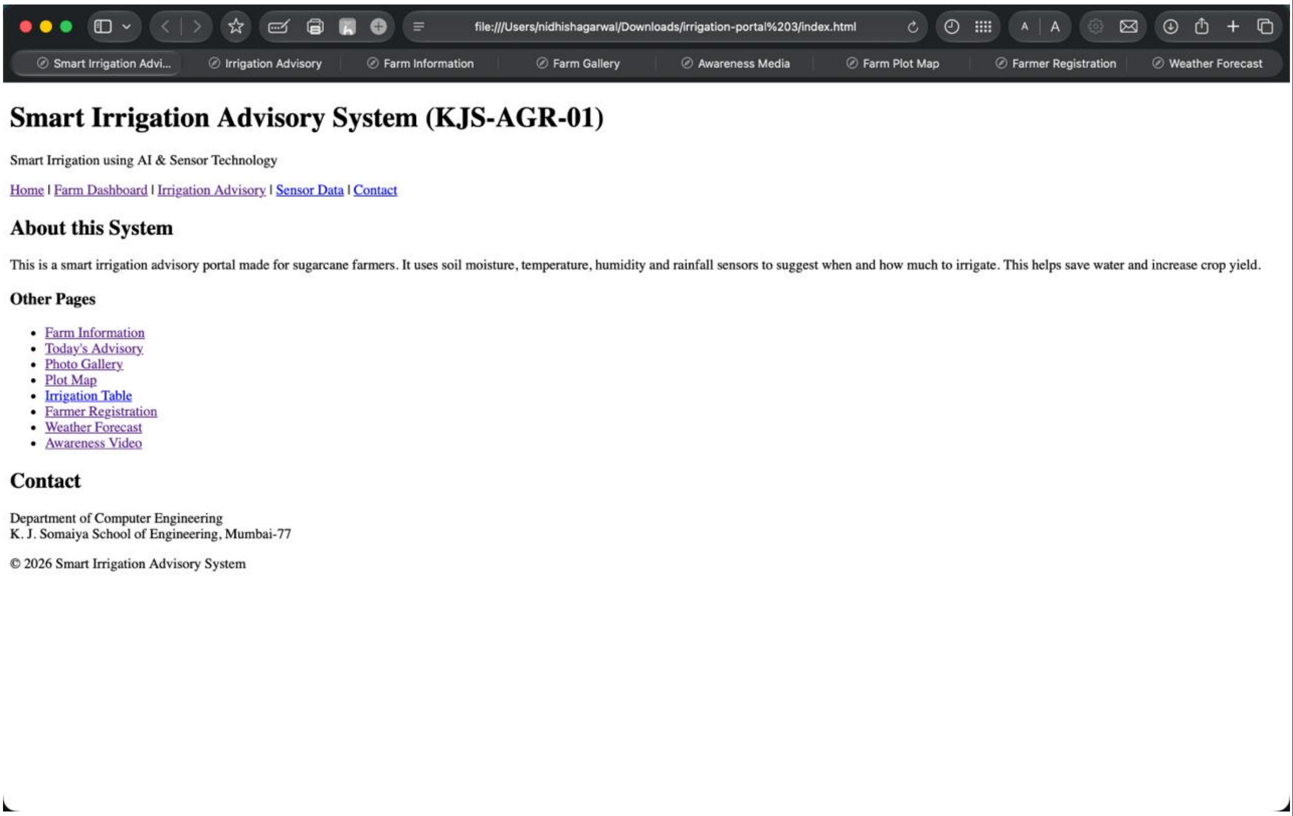
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Awareness Media</title>
</head>
<body>


<header>
  <h1>Smart Irrigation Advisory System (KJS-AGR-01)</h1>
  <p>Smart Irrigation using AI &amp; Sensor Technology</p>
</header>


<nav>
  <a href="index.html">Home</a> |
  <a href="farm-info.html">Farm Dashboard</a> |
  <a href="advisory.html">Irrigation Advisory</a> |
```

```
  <a href="irrigation-table.html">Sensor Data</a> |
  <a href="index.html#contact">Contact</a>
</nav>


<section>
  <h2>Smart Irrigation Awareness Video</h2>
  <video width="480" height="270" controls>
    <source src="media/smart-irrigation.mp4" type="video/mp4">
    Your browser does not support the video tag.
  </video>
</section>


<section>
  <h2>Water Conservation Audio</h2>
  <audio controls loop>
    <source src="audio/water-conservation.mp3" type="audio/mpeg">
    Your browser does not support the audio tag.
  </audio>
</section>


<footer>
  <p>&copy; 2026 Smart Irrigation Advisory System</p>
</footer>
<script>
  window.onload = function () {
    alert("Welcome to the Irrigation Advisory System");
  };
```

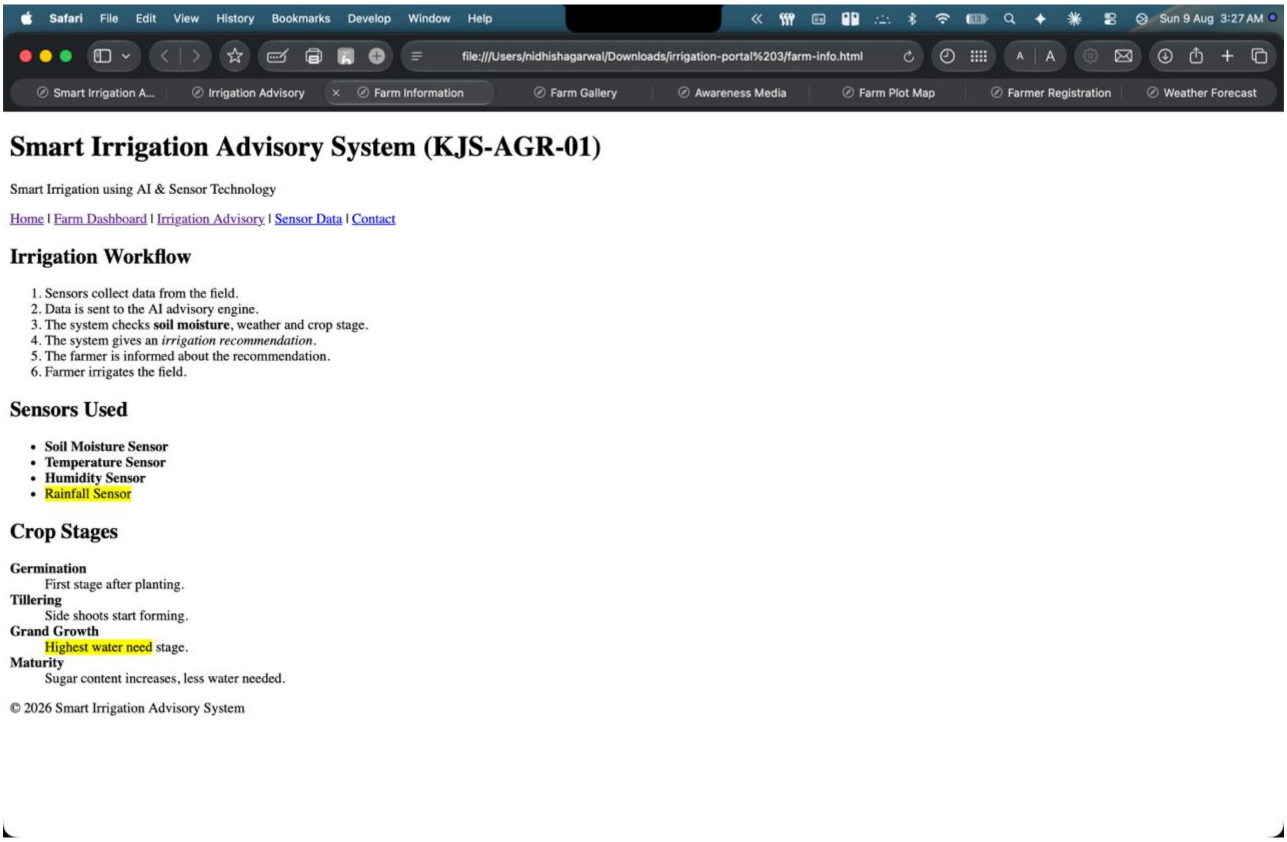
```
</script>
</body>
</html>
```

Output / Screenshot:

Task 1



Task 2



Smart Irrigation Advisory System (KJS-AGR-01)

Smart Irrigation using AI & Sensor Technology

Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact

Irrigation Workflow

1. Sensors collect data from the field.
2. Data is sent to the AI advisory engine.
3. The system checks **soil moisture**, weather and crop stage.
4. The system gives an *irrigation recommendation*.
5. The farmer is informed about the recommendation.
6. Farmer irrigates the field.

Sensors Used

- **Soil Moisture Sensor**
- **Temperature Sensor**
- **Humidity Sensor**
- Rainfall Sensor

Crop Stages

Germination
First stage after planting.
Tillering
Side shoots start forming.
Grand Growth
Highest water need stage.
Maturity
Sugar content increases, less water needed.

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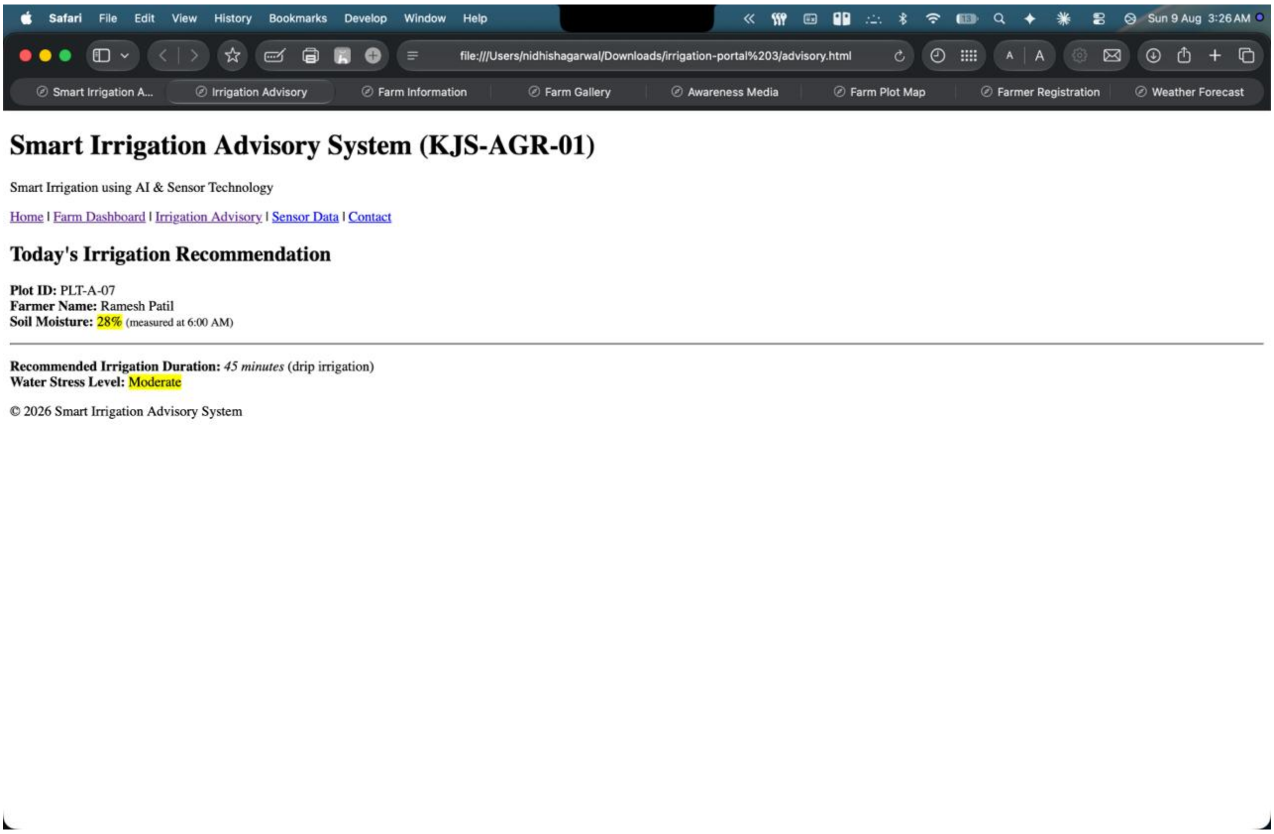
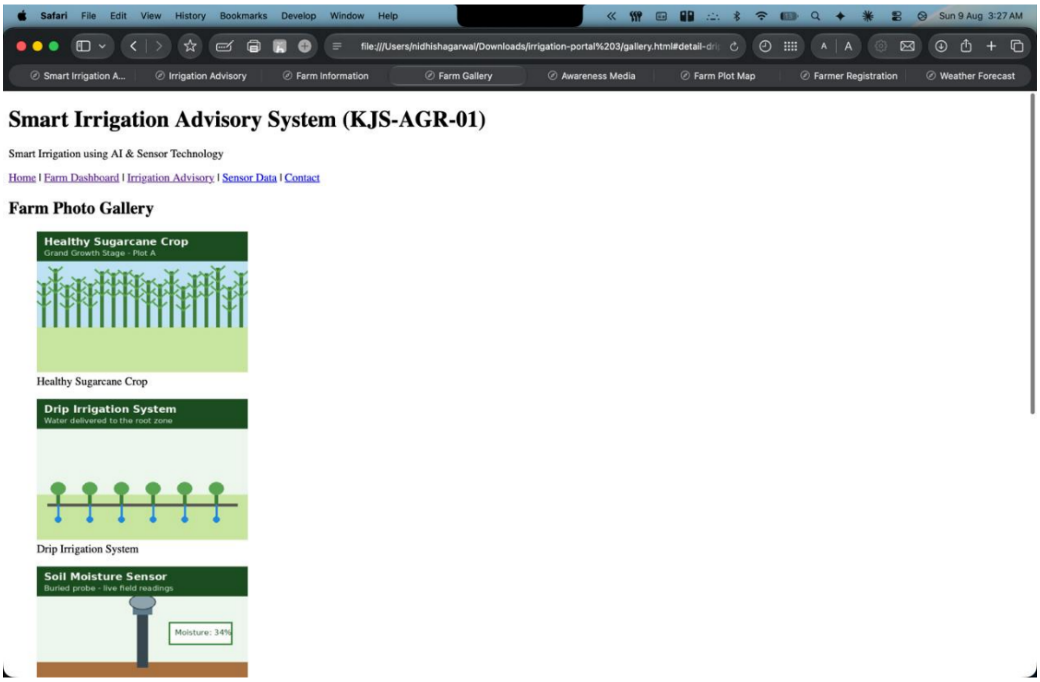
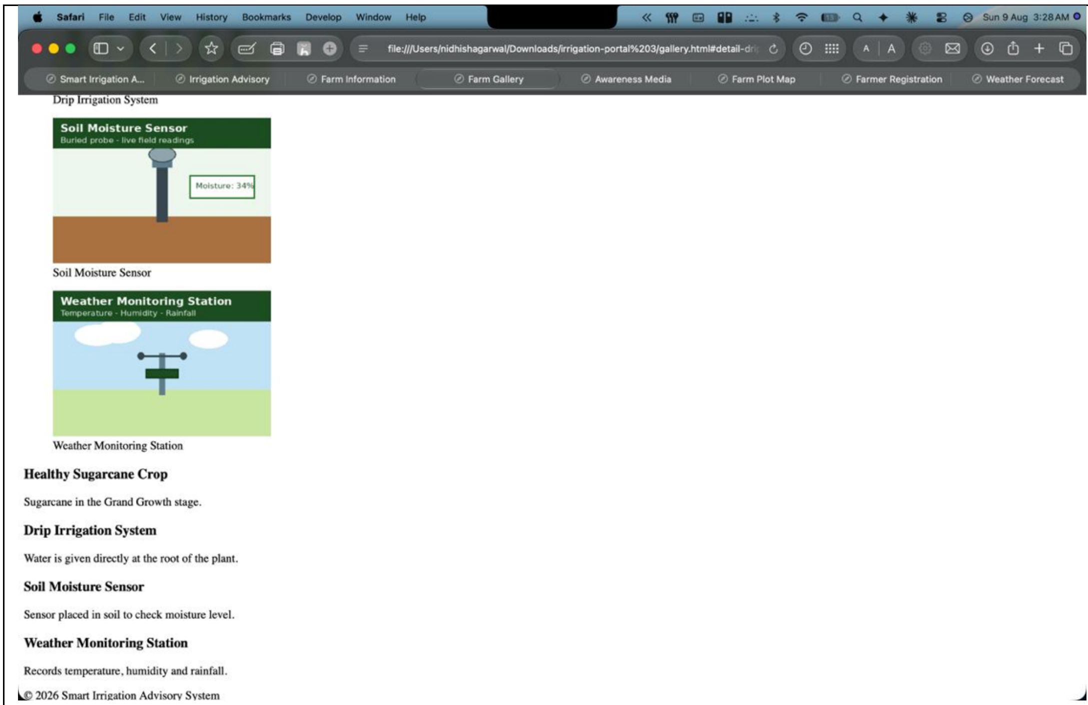
Task 3

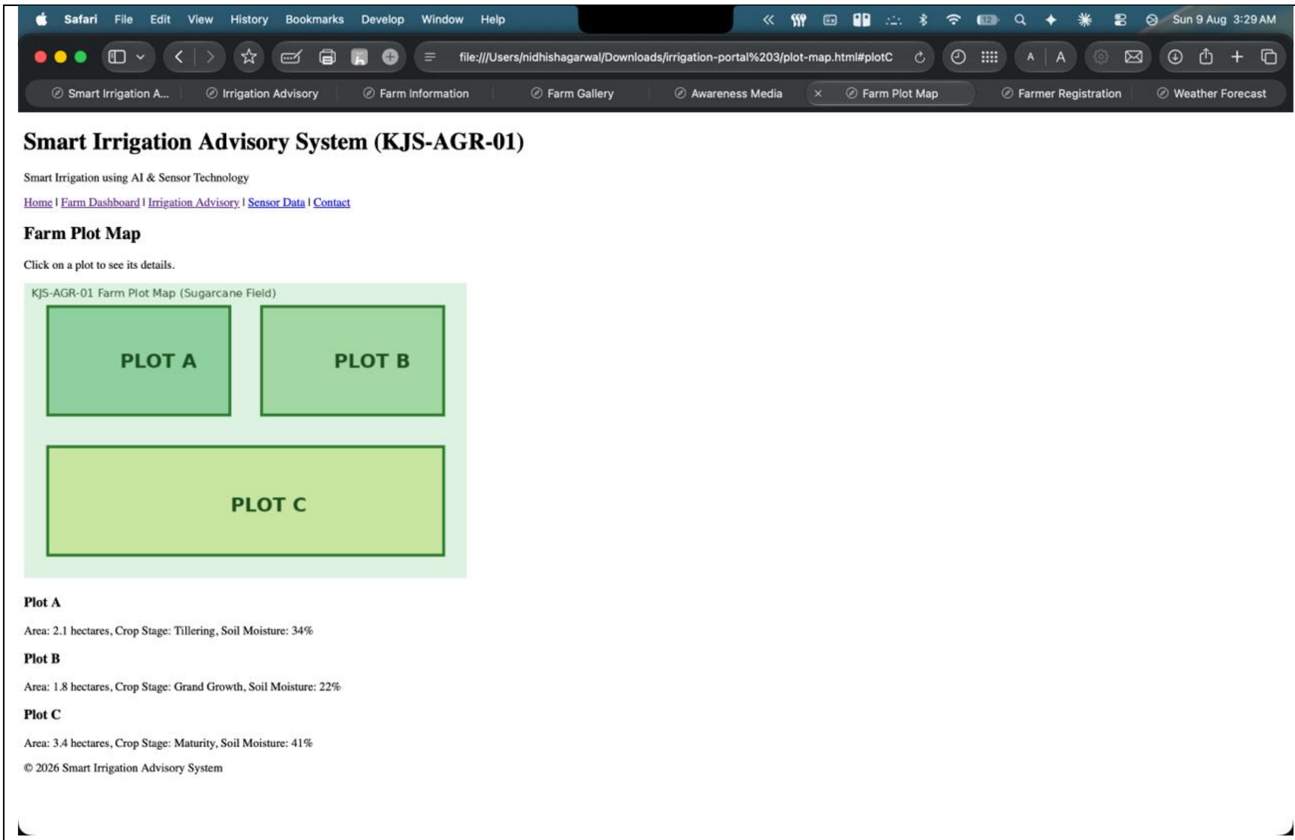

Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:26 AM
file:///Users/nidhishagarwal/Downloads/irrigation-portal%203/advisory.html
Smart Irrigation A...
Irrigation Advisory
Farm Information
Farm Gallery
Awareness Media
Farm Plot Map
Farmer Registration
Weather Forecast
Smart Irrigation Advisory System (KJS-AGR-01)
Smart Irrigation using AI & Sensor Technology
Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact
Today's Irrigation Recommendation
Plot ID: PLT-A-07
Farmer Name: Ramesh Patil
Soil Moisture: 28% (measured at 6:00 AM)
Recommended Irrigation Duration: 45 minutes (drip irrigation)
Water Stress Level: Moderate
© 2026 Smart Irrigation Advisory System

Task 4

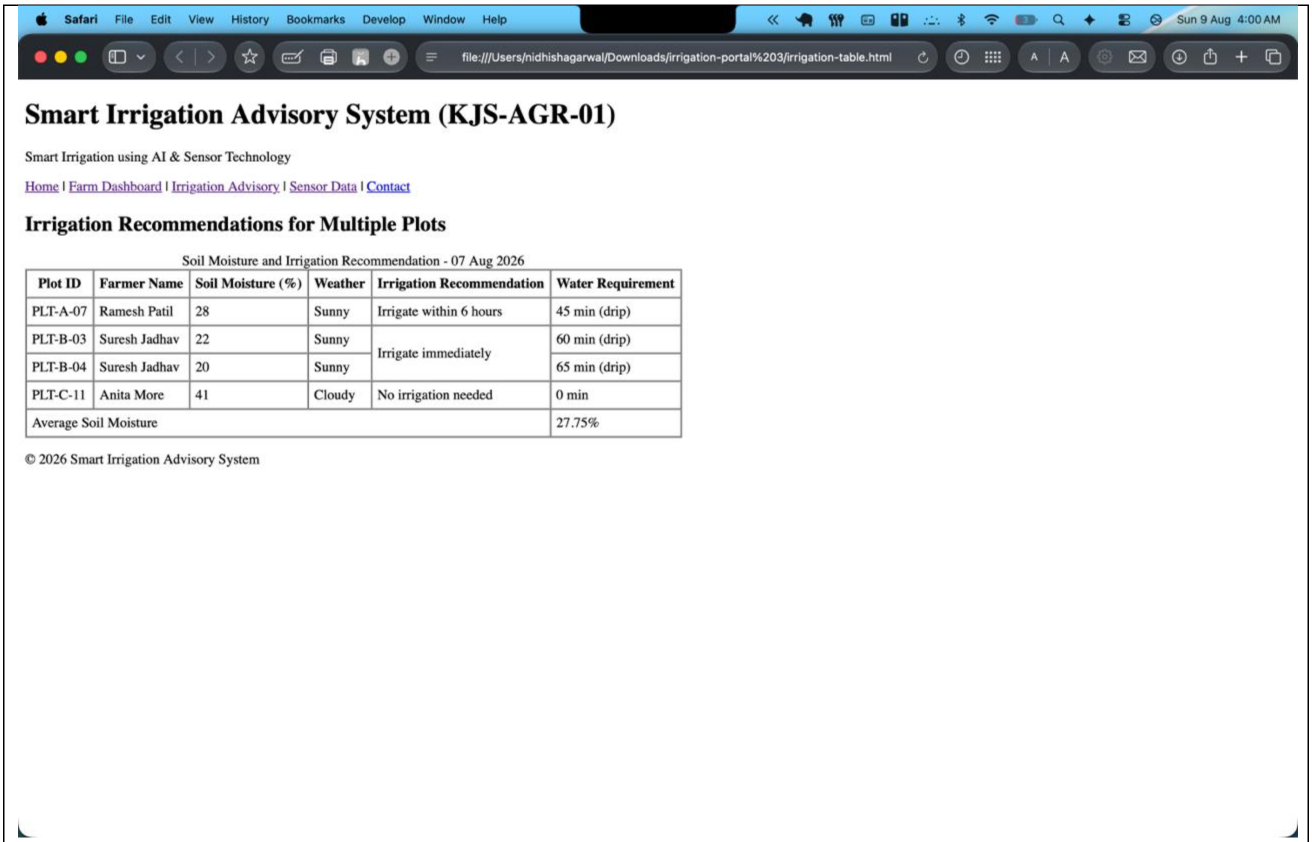

Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:27 AM
Smart Irrigation A...
Irrigation Advisory
Farm Information
Farm Gallery
Awareness Media
Farm Plot Map
Farmer Registration
Weather Forecast
Smart Irrigation Advisory System (KJS-AGR-01)
Smart Irrigation using AI & Sensor Technology
Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact
Farm Photo Gallery
Healthy Sugarcane Crop
Grand Growth Stage · Plot A
Healthy Sugarcane Crop
Drip Irrigation System
Water delivered to the root zone
Drip Irrigation System
Soil Moisture Sensor
Buried probe · live field readings
Moisture: 34%


Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:28 AM
file:///Users/nidhishagarwal/Downloads/irrigation-portal%203/gallery.html#detail-dri
Smart Irrigation A...
Irrigation Advisory
Farm Information
Farm Gallery
Awareness Media
Farm Plot Map
Farmer Registration
Weather Forecast
Drip Irrigation System
Soil Moisture Sensor
Buried probe · live field readings
Moisture: 34%
Soil Moisture Sensor
Weather Monitoring Station
Temperature · Humidity · Rainfall
Weather Monitoring Station
Healthy Sugarcane Crop
Sugarcane in the Grand Growth stage.
Drip Irrigation System
Water is given directly at the root of the plant.
Soil Moisture Sensor
Sensor placed in soil to check moisture level.
Weather Monitoring Station
Records temperature, humidity and rainfall.
© 2026 Smart Irrigation Advisory System

Task 5


Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:29 AM
file:///Users/nidhishagarwal/Downloads/irrigation-portal%203/plot-map.html#plotC
Smart Irrigation A...
Irrigation Advisory
Farm Information
Farm Gallery
Awareness Media
Farm Plot Map
Farmer Registration
Weather Forecast
Smart Irrigation Advisory System (KJS-AGR-01)
Smart Irrigation using AI & Sensor Technology
Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact
Farm Plot Map
Click on a plot to see its details.
KJS-AGR-01 Farm Plot Map (Sugarcane Field)
PLOT A
PLOT B
PLOT C
Plot A
Area: 2.1 hectares, Crop Stage: Tillering, Soil Moisture: 34%
Plot B
Area: 1.8 hectares, Crop Stage: Grand Growth, Soil Moisture: 22%
Plot C
Area: 3.4 hectares, Crop Stage: Maturity, Soil Moisture: 41%
© 2026 Smart Irrigation Advisory System

Task 6



Smart Irrigation Advisory System (KJS-AGR-01)

Smart Irrigation using AI & Sensor Technology

Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact

Irrigation Recommendations for Multiple Plots

Soil Moisture and Irrigation Recommendation - 07 Aug 2026

Plot ID	Farmer Name	Soil Moisture (%)	Weather	Irrigation Recommendation	Water Requirement
PLT-A-07	Ramesh Patil	28	Sunny	Irrigate within 6 hours	45 min (drip)
PLT-B-03	Suresh Jadhav	22	Sunny	Irrigate immediately	60 min (drip)
PLT-B-04	Suresh Jadhav	20	Sunny		65 min (drip)
PLT-C-11	Anita More	41	Cloudy	No irrigation needed	0 min
Average Soil Moisture					27.75%

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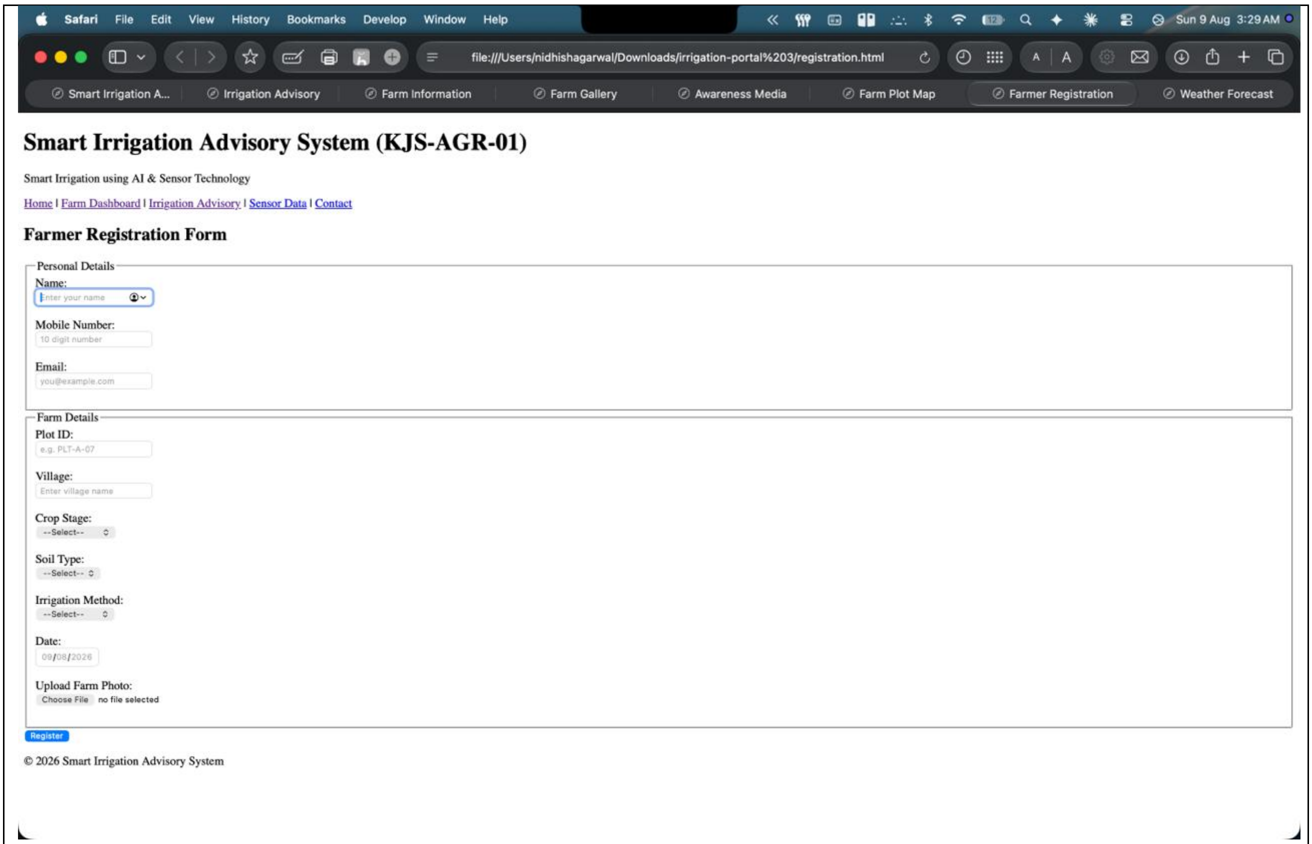
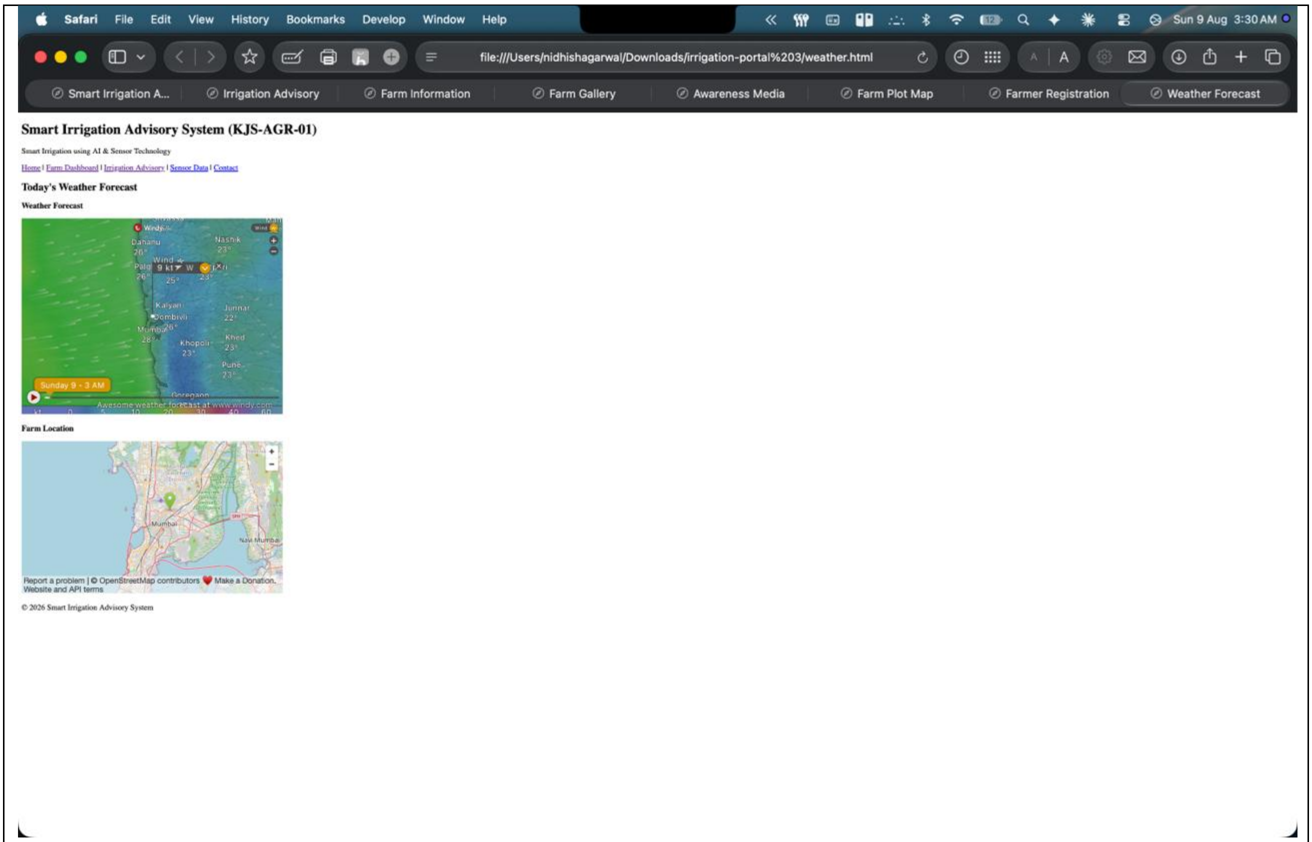
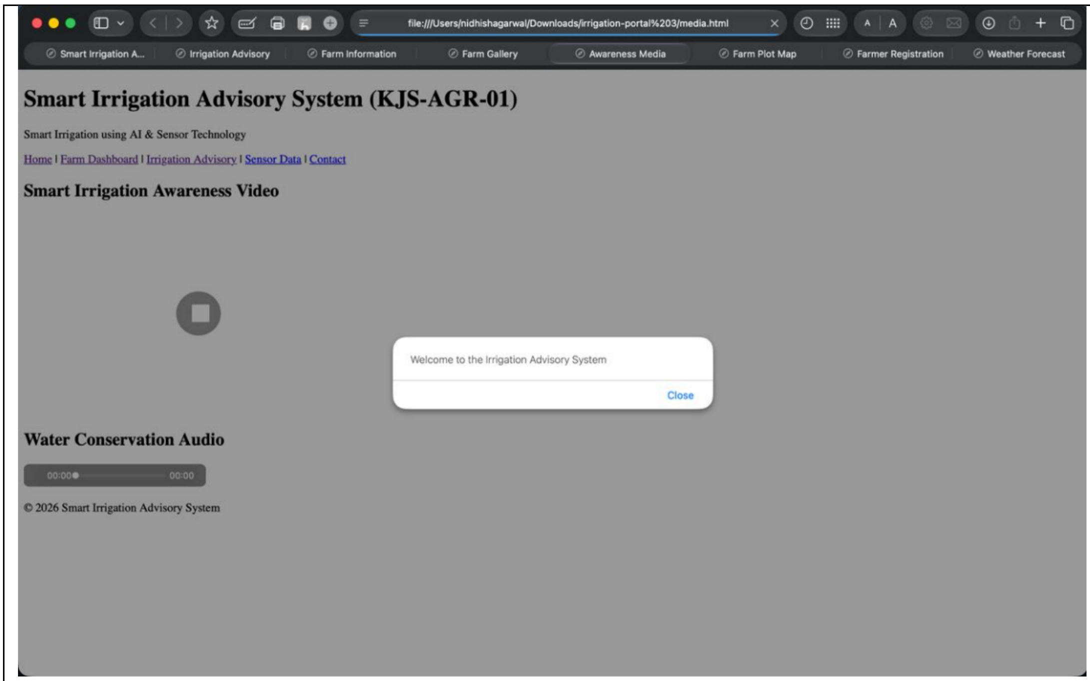
Task 7

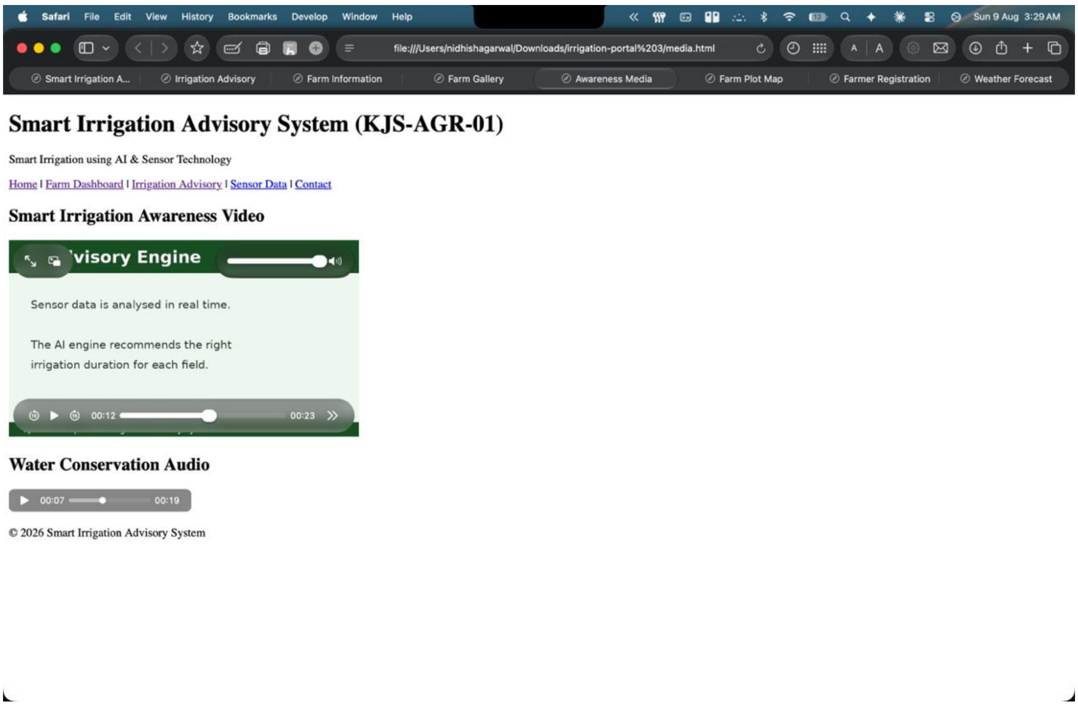

Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:29 AM
file:///Users/nidhishagarwal/Downloads/irrigation-portal%203/registration.html
Smart Irrigation A... Irrigation Advisory Farm Information Farm Gallery Awareness Media Farm Plot Map Farmer Registration Weather Forecast
Smart Irrigation Advisory System (KJS-AGR-01)
Smart Irrigation using AI & Sensor Technology
Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact
Farmer Registration Form
Personal Details
Name:
Enter your name
Mobile Number:
10 digit number
Email:
you@example.com
Farm Details
Plot ID:
e.g. PLT-A-07
Village:
Enter village name
Crop Stage:
--Select--
Soil Type:
--Select--
Irrigation Method:
--Select--
Date:
09/08/2026
Upload Farm Photo:
Choose File no file selected
Register
© 2026 Smart Irrigation Advisory System

Task 8


Safari File Edit View History Bookmarks Develop Window Help
Sun 9 Aug 3:30 AM
file:///Users/nidhishagarwal/Downloads/irrigation-portal%203/weather.html
Smart Irrigation A...
Irrigation Advisory
Farm Information
Farm Gallery
Awareness Media
Farm Plot Map
Farmer Registration
Weather Forecast
Smart Irrigation Advisory System (KJS-AGR-01)
Smart Irrigation using AI & Sensor Technology
Home | Farm Dashboard | Irrigation Advisory | Sensor Data | Contact
Today's Weather Forecast
Weather Forecast
Farm Location
© 2026 Smart Irrigation Advisory System

Task 9





Conclusion and Discussion:

In this experiment, I successfully built a Smart Irrigation Advisory System Portal using fundamental HTML5 concepts. Working through all 9 tasks helped me practically understand how to build web pages using semantic tags, organize data with lists and tables, create interactive forms with validation, and set up image maps. I also learned how to embed external sites using iframes and add multimedia like audio, video, and basic JavaScript. Overall, this lab gave me hands-on experience in using web development to solve real world farming challenges.